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|-----------------------------------------------------------------|---------------------|---------------------------------------------------------|---------------------|
| <b>FACULTY/COLLEGE</b>                                          |                     | College of Business and Economics                       |                     |
| <b>SCHOOL</b>                                                   |                     | School of Consumer Intelligence and Information Systems |                     |
| <b>DEPARTMENT</b>                                               |                     | Applied Information Systems                             |                     |
| <b>CAMPUS(ES)</b>                                               |                     | APB                                                     |                     |
| <b>MODULE NAME</b>                                              |                     | Development Software 2A                                 |                     |
| <b>MODULE CODE</b>                                              |                     | DSW02A1                                                 |                     |
| <b>SEMESTER</b>                                                 |                     | First                                                   |                     |
| <b>ASSESSMENT OPPORTUNITY, MONTH AND YEAR</b>                   |                     | Graded Lab Assessment<br>May 2025                       |                     |
| <b>ASSESSMENT DATE</b>                                          | 24 May 2025 @ 09:00 | <b>SUBMISSION DATE</b>                                  | 24 May 2025 @ 23h59 |
| <b>ASSESSOR(S)</b>                                              | Mr Ronny Mabokela   |                                                         |                     |
|                                                                 |                     |                                                         |                     |
| <b>DURATION</b>                                                 | 8 hours             | <b>TOTAL MARKS</b>                                      | 130                 |
| <b>NUMBER OF PAGES OF QUESTION PAPER (Including cover page)</b> |                     |                                                         | 8                   |

**INFORMATION/INSTRUCTIONS:**

- ✓ Read the instructions carefully to avoid making mistakes.
- ✓ This is NOT an open-book assessment.
- ✓ There are 5 questions and answer all questions.
- ✓ Submit the necessary files for codes where necessary.
- ✓ Do not provide **ChatGPT** code, if detected as AI-generated, you will be given Zero as a mark.
- ✓ **DO NOT SHARE THE QUESTION PAPER IN THE WHATSAPP GROUP**
- ✓ **EACH STUDENT MUST ACCESS THE QUESTION PAPER FROM THE BLACKBOARD**

**QUESTION 1****[20 MARKS]**

Write the PHP code for a web page **campdonations.php** that displays donors who have donated amounts in a specified range during the South African Presidential Candidate Campaign in 2017. Assume that there is a provided page with a form where the user can type the name of a donor and select a donation level. The form will submit to your campdonations.php. Here is the relevant HTML from that provided page, and a screenshot of its appearance:

```
<h1>Donation Search</h1>
<form action="donations.php" method="get">
<div>
<input type="radio" name="level" value="bronze"/> bronze
<input type="radio" name="level" value="silver"/> silver
<input type="radio" name="level" value="gold"/> gold
</div>
<div><input type="text" name="donor" /> Donor name</div>
<div><input type="submit" value="View donations" /></div>
</form>
```

Each donor has a corresponding text file in the current directory that contains all donations for that donor.

- The file may not have the exact name of the donor but it will contain the exact name.
- For example, if the user types “CR17” it should match “cr17forpresident.txt” or “NDZ17” it should match from “ndz17forpresident.txt”.
- If there is more than one file that contains the name of the donors you should select the first one.
- Each line in the file contains the donator’s name, the donation amount and “yes” if the donation was matched, and “no” otherwise.
- Each item is separated from the others by a semicolon (“.”), as in the example below at right.

The bronze donation range amounts between 0m and 1m inclusive, silver between 2m exclusive and 5m inclusive and gold above 5m and 10m inclusive.

- Your PHP page should accept the query parameters from the above form, open the text file for the correct donor, and search it for donations in the range matching the selected category.
- Display each donation as a bullet in an unordered list, showing the donor and the amount separated by a dash (-). At the bottom of the page output the total amount donated from this group.
- If the donation was matched, the amount donated should be added to the total twice.

CR17 for president: Donors

|                                               |
|-----------------------------------------------|
| De Bears: R5m:no<br>Patrice Motsepe: R2m: yes |
|-----------------------------------------------|

Nicky Oppenheimer: R10m: yes  
 Sifiso Dabengwa: R1m: yes  
 Maria Ramos: R10m: yes  
 Colin Coleman: R3m: yes  
 Mark Lambert: R5m: yes  
 Raymond Ackerman: R1m: yes  
 Julius Malema: R2m: no

NDZ17 for president: Donors

Crony and friend: R1m: yes  
 Ace Magashule: R2m: yes  
 Guptas: R10m: yes  
 Tom Moyani: R1m: yes  
 Jacob Zuma: R5m: yes  
 DSW Students: R500: no  
 Zuma and Sons: R5m: no

You may assume that the user will submit parameters with the appropriate values (e.g. the level will always be gold, silver or bronze) if they submit parameters. However, they may omit parameters.

- If they do your code should print an error message explaining what went wrong.
- You may assume that a file for the donor the user submits exists and is valid in the format described above.
- You can write just the code that would go inside the page body; you do not need it.

The following screenshots show the PHP page output after the user submits the form with various values:

|                                                                                                                                                                                               |                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><u>1. Bronze – donor: CR17 for president</u></p> <ul style="list-style-type: none"> <li>• Raymond Ackerman - 1m</li> <li>• Sifiso Dabengwa – 1m</li> </ul> <p><b><u>Total: R4m</u></b></p> | <p><u>2. Silver – donor: CR17 for president</u></p> <ul style="list-style-type: none"> <li>• Patrice Motsepe - R2m</li> <li>• Colin Coleman - R3m</li> <li>• Mark Lambert - R5m</li> </ul> <p><b><u>Total: R20m</u></b></p> |
| <p><u>2. Bronze – donor: NDZ for president</u></p> <ul style="list-style-type: none"> <li>• Crony and friends - R1m</li> <li>• Tom Moyani - R1m</li> </ul> <p><b><u>Total: R4m</u></b></p>    | <p><u>3. No level selected</u></p> <p>Please, specify both the donor name and category level</p>                                                                                                                            |

## QUESTION 5

[10 MARKS]

Use the skills learned this semester to write the JavaScript code to accompany the following HTML code so that when the Delete button is clicked by a user, any button whose text value

is divisible by the number written in the text field is removed from the page. You can choose any number that is positive (+1, + infinity).

The behaviour should be as follows:

- You may assume that a valid number has been typed in the text field.
- Zero cannot be used as an option for the program.
- No negative number must be allowed in the system.
- Style your HTML with a good color of your choice.
- Add the following to the given values: 120, 133, 180, 203 204, 220, 226, 227, 230.
- Make sure that once you refresh your code can still work with all buttons.

**Use the HTML code below:**

```
<div id="controls">Divisible by: <input type="text" id="divisor" />
<button id="del">Delete</button>
</div>
<div id="buttons">
  Click a button:
  <button>11</button> <button>22</button>
  <button>34</button> <button>42</button>
  <button>50</button> <button>63</button>
  <button>71</button> <button>85</button>
  <button>94</button> <button>103</button>
</div>
```

Example: The picture below shows the following of the initial page and its new appearance after typing 2 into the text field and pressing the button Delete. Thus display the values that are deleted.

